

Note 1.2 Combining Functions

Sums, Differences, Products and Quotients

- **Formula:**
 - $(f \pm g)(x) = f(x) \pm g(x), x \in D(f) \cap D(g)$
 - $(fg)(x) = f(x)g(x), x \in D(f) \cap D(g)$
 - $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, x \in D(f) \cap D(g) \text{ and } g(x) \neq 0$

Composite Functions

- **Definition:** the composite function $f \circ g$ is defined by

$$(f \circ g)(x) = f(g(x))$$

and $x \in D(g), R(g)$

Shifting a Graph of a Function

- **Shift Formula:**
Vertical shift: $y = f(x) + k$, shift up k units if $k > 0$ and shift down $|k|$ units if $k < 0$
Horizontal shift: $y = f(x + h)$, shift left h units if $h > 0$ and shift right if $h < 0$

Scaling and Reflecting a Graph of a Function

- **Formulas:**
For $c > 1$, the graph is scaled:
 $y = cf(x)$, stretch vertically by c
 $y = \frac{1}{c}f(x)$, compress vertically by c
 $y = f(cx)$, compress horizontally by c
 $y = f\left(\frac{x}{c}\right)$, stretch horizontally by c
For $c = -1$, the graph is reflected:
 $y = -f(x)$, reflect across the x -axis
 $y = f(-x)$, reflect across the y -axis

Ellipses

- **Process to get an ellipse:**
 - circle: $x^2 + y^2 = r^2$
 - stretch or compress the circle by c to get an ellipse: $c^2x^2 + y^2 = r^2$

- Quotient c^2 of both sides:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

in which $a = \frac{x}{c}, b = r$.